

DIGITAL TIME CAPSULE

IT23431 –MONGODB PROJECT REPORT

Submitted by

K SNEHITHA– 241001254

M SANTHIYA-241001228

OF

BACHELOR OF TECHNOLOGY IN INFORMATION TECHNOLOGY

RAJALAKSHMI ENGINEERING COLLEGE, THANDALAM

(An Autonomous Institution)



RAJALAKSHMI ENGINEERING COLLEGE

BONAFIDE CERTIFICATE

Certified that this project titled “Digital Time Capsule” is the bonafide work of
“K SNEHITHA (241001254), M SANTHIYA(241001228)” who carried out the
project work under my supervision.

SIGNATURE

Dr.Valarmathie P

HEAD OF THE DEPARTMENT

Information Technology

Rajalakshmi Engineering College,

Rajalakshmi Nagar, Thandalam

Chennai – 602105

SIGNATURE

Mrs.Usha S

COURSE INCHARGE

Information Technology

Rajalakshmi Engineering College,

Rajalakshmi Nagar, Thandalam

Chennai - 602105

This project is submitted for IT23431 –MongoDb Essentials held on _____

INTERNAL EXAMINAR

EXTERNAL EXAMINAR

TABLE OF CONTENTS

S.No	Title	Page No
1.1	Abstract	4
1.2	Introduction	4
1.3	Purpose	4
1.4	Scope of the project	5
1.5	Software requirement specification	5
2.1	Use case diagram	6
2.2	Data flow diagram	6
2.3	Entity-relationship diagram	7
3	Module description	8
4	Implementation	9
5	Conclusion	12
6	References	12

1 DIGITAL TIME CAPSULE

1.1 ABSTRACT

The Digital Time Capsule System is a web-based application designed to securely store digital content such as messages, images, and documents, which can be accessed at a specified future date. The system uses modern technologies like Node.js for backend processing and MongoDB as the database for efficient and scalable data storage.

The main objective of this system is to provide a platform where users can preserve memories or important information and retrieve them in the future. The integration of MongoDB ensures fast data retrieval, flexibility, and secure storage of unstructured data. With a user-friendly interface and time-lock mechanism, this project demonstrates how digital technology can be used for long-term data preservation in a secure and efficient way.

1.2 INTRODUCTION

The Digital Time Capsule System is designed to allow users to store digital data and access it after a specific period. Unlike traditional storage methods, this system ensures that the stored content remains hidden until the chosen unlock date. It manages tasks such as user registration, content uploading, and timed access control. The system reduces the risk of data loss and provides a secure way to store personal or important information. With database support and secure authentication, users can easily manage their digital capsules. This project aims to provide a reliable, secure, and efficient solution for future data access and memory preservation.

1.3 PURPOSE

The purpose of the Digital Time Capsule System is to:

- Provide a secure platform for storing digital memories
- Enable users to send messages to their future selves
- Reduce the risk of data loss
- Ensure privacy and controlled access using time-lock features
- Improve digital data management and storage efficiency

1.3 SCOPE OF THE PROJECT

This project includes functionalities such as user registration, secure login, uploading digital content, setting unlock dates, and retrieving data after the specified time.

It can be used by individuals, students, or organizations for storing memories, important documents, or messages for future use. The system can be expanded with features like cloud storage, email notifications, and sharing options, making it scalable and adaptable.

1.4 FUNCTIONAL REQUIREMENTS

1. **User Authentication Module (UAM)**

Users can register and log in securely using username and password.
Authentication is verified through the database.

2. **Capsule Creation Module (CCM)**

Users can create a new time capsule by uploading text, images, or files.
Users set a future date for unlocking the capsule.

3. **Capsule Management Module (CMM)**

Users can view, update, or delete capsules before the unlock date.
Capsules remain locked until the specified date.

4. **Time-Lock Module (TLM)**

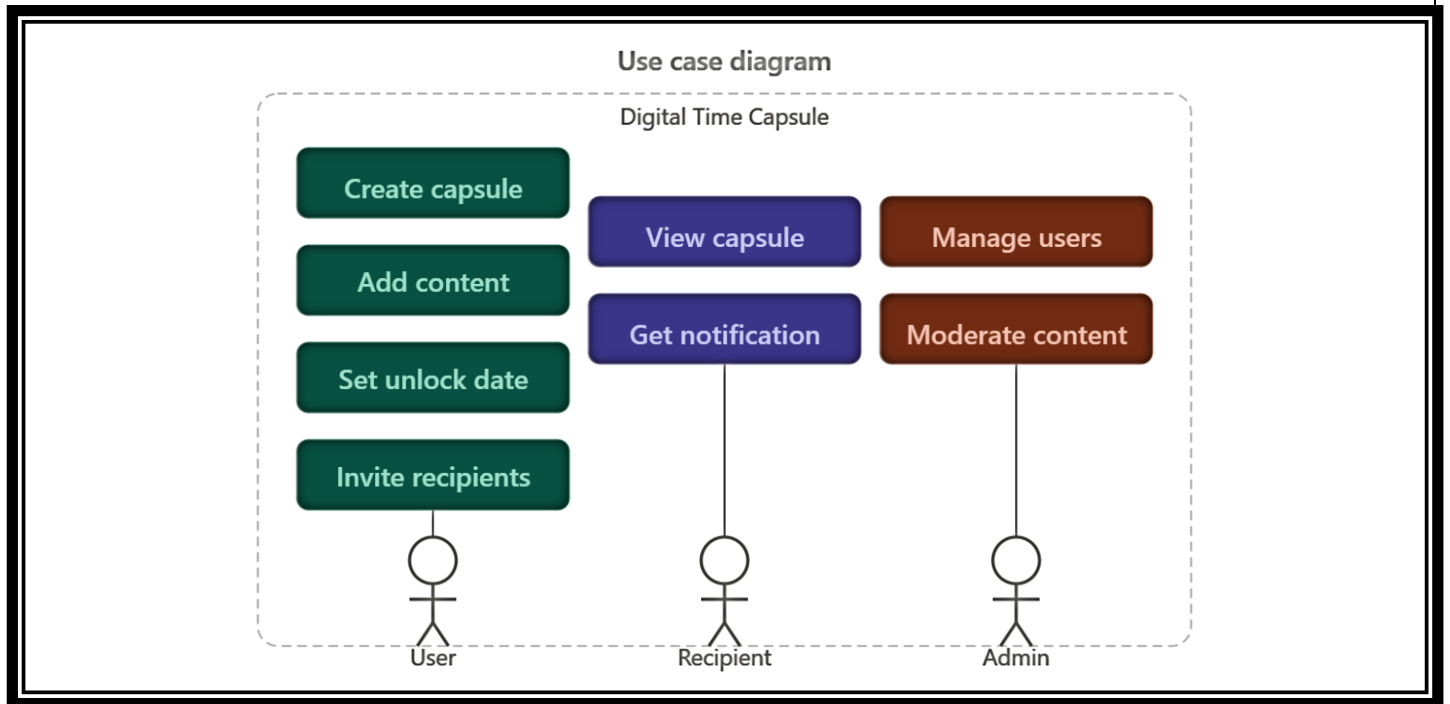
Ensures that stored content cannot be accessed before the unlock date.
Automatically unlocks data when the date is reached.

5. **Notification Module (NM)**

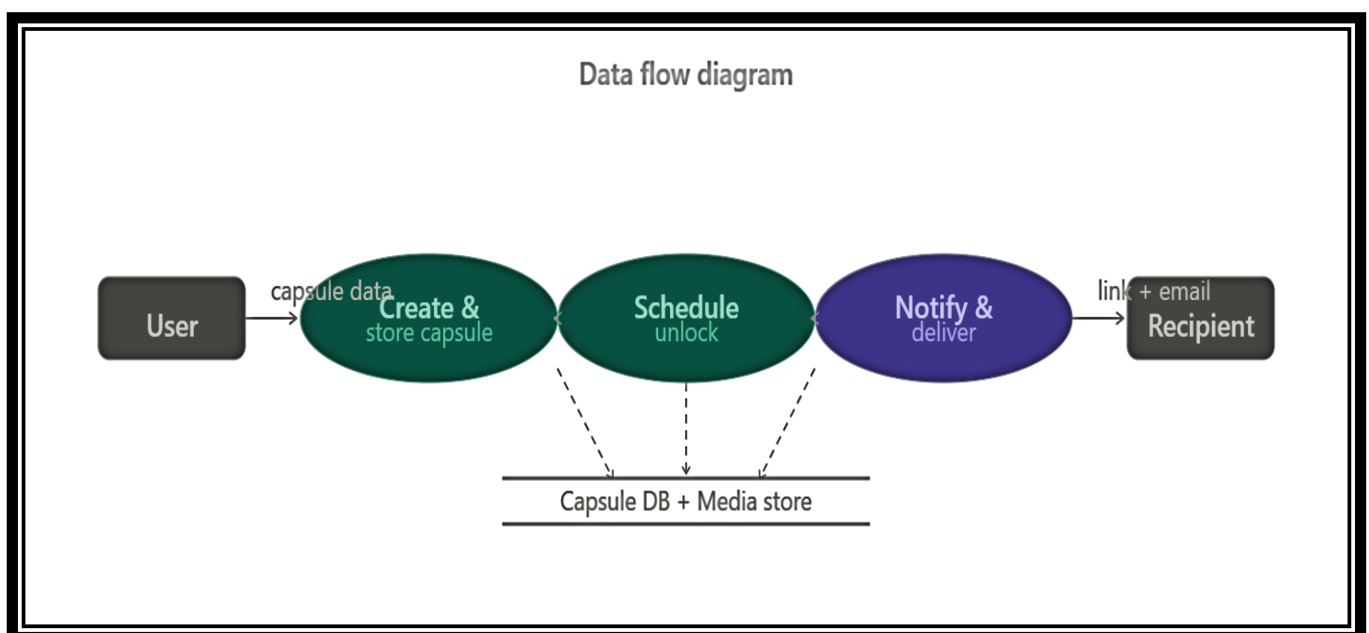
Users receive notifications when their capsule becomes accessible.

2.SYSTEM FLOW DIAGRAMS

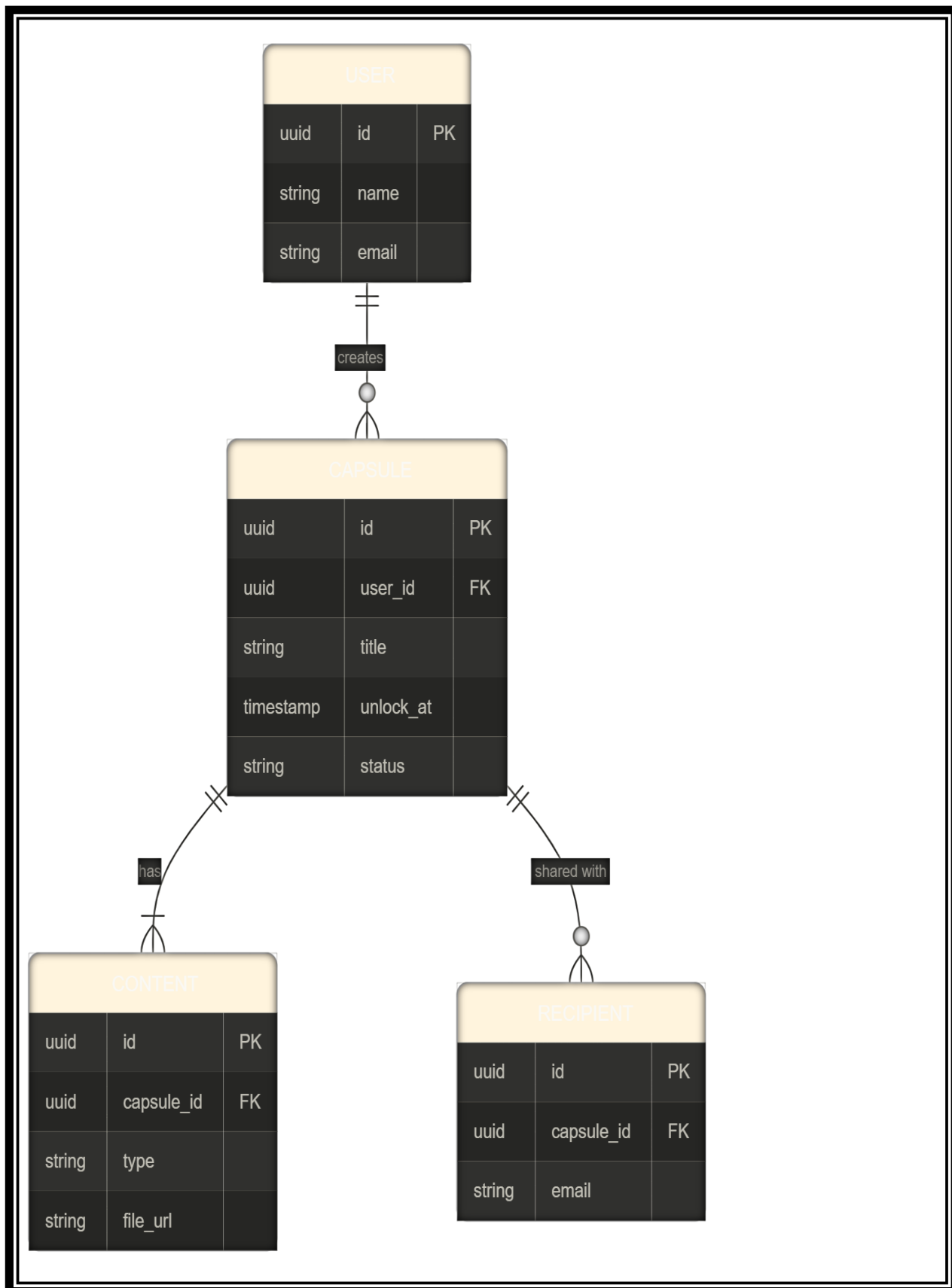
2.1 USE CASE DIAGRAMS



2.2 DATA FLOW DIAGRAMS



2.3 ENTITY RELATIONSHIP DIAGRAMS



3.MODULE DESCRIPTION

USER LOGIN

Users log in using their credentials. Access is granted only after verification.

CREATE CAPSULE

Users can create a capsule by adding messages, images, or files and setting an unlock date.

MANAGE CAPSULES

Users can view or edit capsules before the unlock date.

TIME-LOCK ACCESS

The system restricts access until the specified date is reached.

VIEW CAPSULE

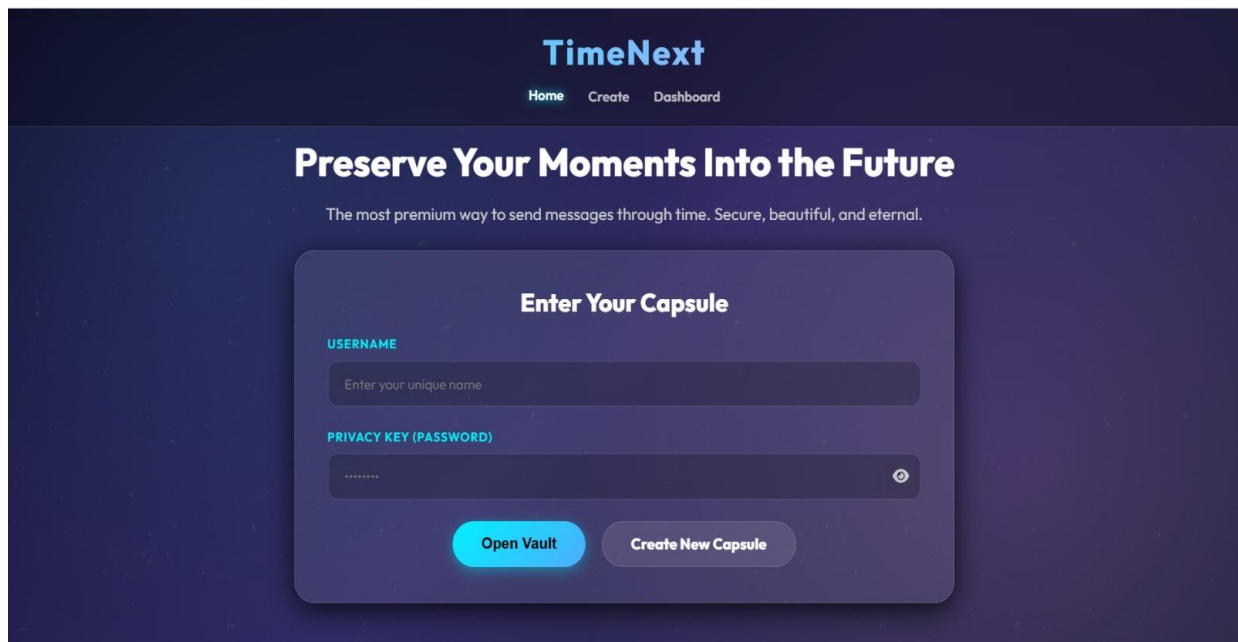
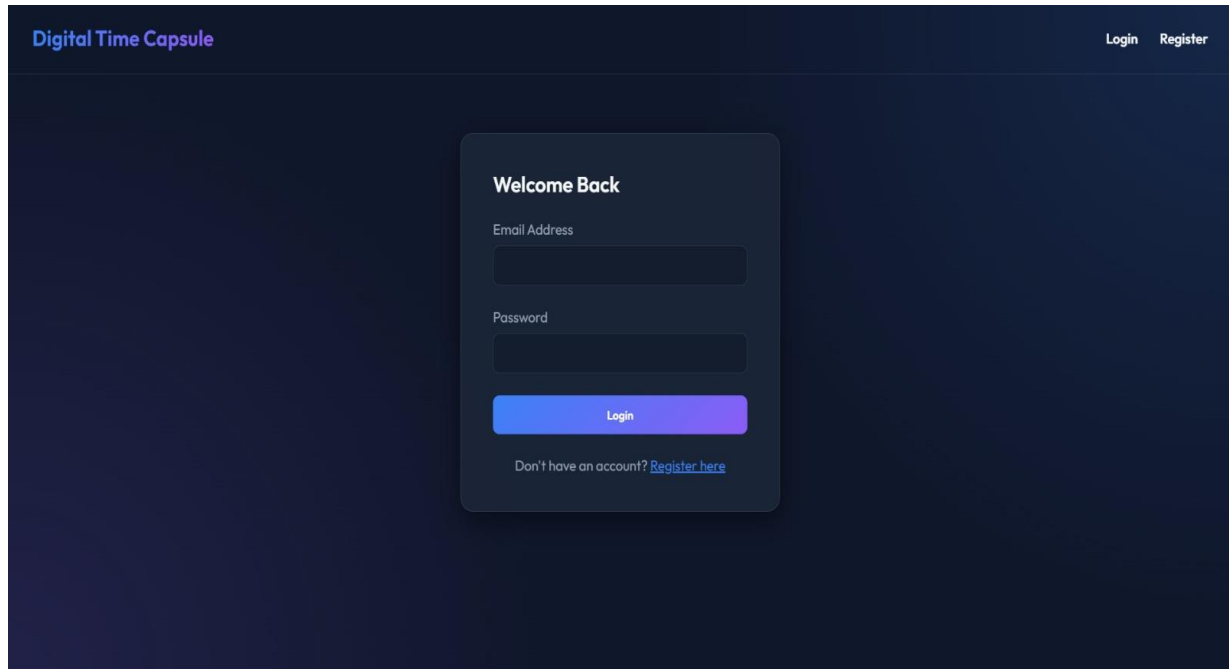
After the unlock date, users can view or download their stored content.

LOGOUT

Users can securely log out after completing their actions.

3.IMPLEMENTATION

3.1 SCREENSHOT



TimeNext

[Home](#)[Create](#)[Dashboard](#)

Create Your Time Capsule

YOUR USERNAME

Who is sending this?

PRIVACY KEY (RECTIFICATION PASSWORD)

Required for future edits

Keep this safe. You'll need it to rectify any mistakes later.

RECIPIENT NAME

Someone special (Optional)

OPENING DATE & TIME (IST)

dd-mm-yyyy

Hour

Min

Period

in Indian Standard Time (IST)

A SPECIAL MEMORY (PHOTO)

Choose File

No file chosen

Pick a photo from your gallery or files.

YOUR MESSAGE TO THE FUTURE

If you could tell yourself or someone else something years from now, what would it be?

Seal Eternal Capsule

TimeNext

[Home](#)[Create](#)[Dashboard](#)

Your Capsules

ENTER YOUR USERNAME TO VIEW YOUR VAULT:

Harini

View Vault

AVAILABLE

To: Harini

Scheduled for 2/21/2026 at 09:14 PM

Open

Delete

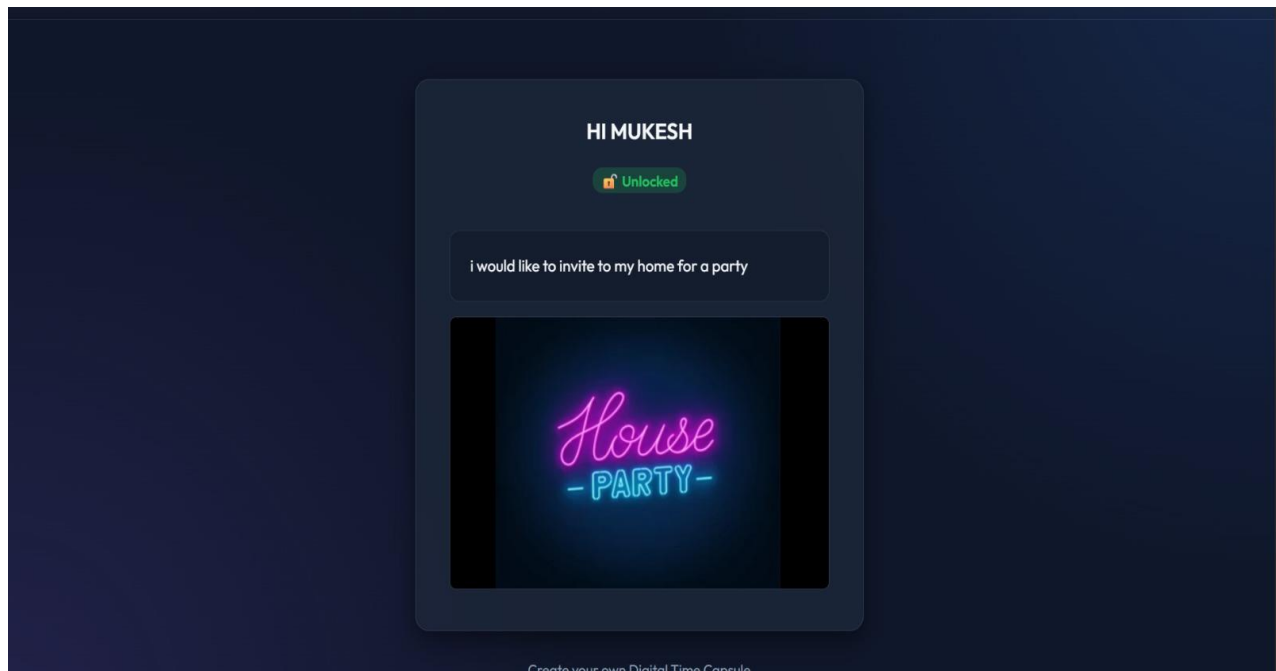
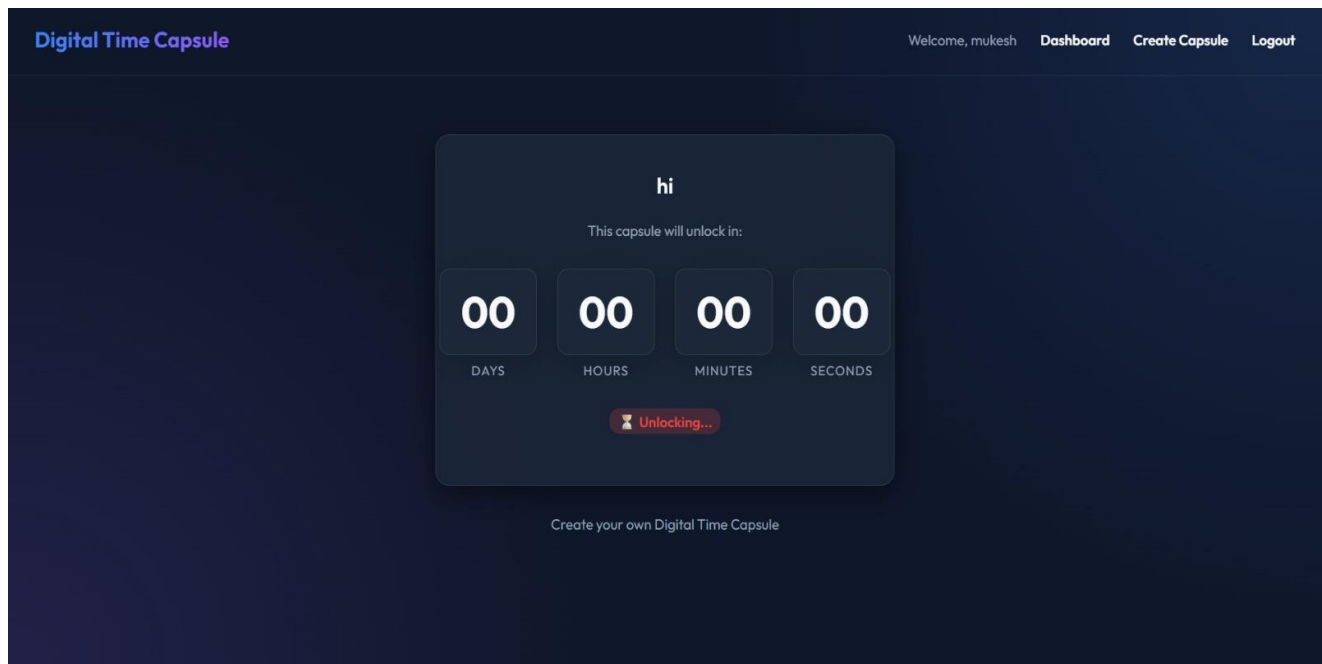
AVAILABLE

To: To myself

Scheduled for 2/23/2026 at 08:15 PM

Open

Delete



4.2 DATA BASE DESIGN

The Digital Time Capsule System uses MongoDB, a NoSQL database, to store and manage data efficiently. It stores data in JSON-like (BSON) format, making it flexible for handling text, images, and files. The main objective of the database is to ensure secure storage, fast retrieval, minimal redundancy, and data consistency.

✓ Collections Used

1. Users

Stores user details.

Fields: `_id`, `username`, `email`, `password`, `createdAt`

2. Capsules

Stores time capsule data.

Fields: `_id`, `userId`, `title`, `message`, `mediaUrls`, `unlockDate`, `isLocked`

3. Files (Optional)

Stores file-related information.

Fields: `_id`, `capsuleId`, `fileName`, `filePath`

4. Notifications

Stores alerts for users.

Fields: `_id`, `userId`, `capsuleId`, `message`, `isRead`

✓ Relationships

- One user can create many capsules
- One capsule can have multiple files

✓ Key Features

- Password encryption for security
- Indexing for faster search (`userId`, `email`)
- Data validation to ensure accuracy
- Time-based access using unlock date

3.3CODE

Connectivity code

```
const express = require("express");
const mongoose = require("mongoose");

const app = express();
app.use(express.json());

// Routes
app.use("/users", require("./routes/userRoutes"));
app.use("/capsules", require("./routes/capsuleRoutes"));

// MongoDB Connection
mongoose.connect("mongodb://127.0.0.1:27017/timeCapsuleDB")
.then(() => console.log("MongoDB Connected"))
.catch(err => console.log(err));

// Test Route
app.get("/", (req, res) => {
  res.send("Digital Time Capsule Server Running");
});

// Server Start
app.listen(5000, () => {
  console.log("Server started on port 5000");
});
```

4.CONCLUSION

The Digital Time Capsule System successfully provides a secure and efficient platform for storing and retrieving digital data in the future. By integrating modern technologies like Node.js and MongoDB, the system ensures scalability, flexibility, and data security.

It reduces manual effort, protects sensitive information, and introduces an innovative approach to memory preservation. Overall, this project demonstrates how technology can be used to store valuable moments and access them at the right time.

4.REFERENCES

- W3Schools – Node.js and MongoDB Tutorials
- GeeksforGeeks – MongoDB Concepts
- TutorialsPoint – Web Development and Database Integration

